1. **Topic 8: Object-Oriented Modeling (The OOP Choice)**

   Or Fundamentals of programming languages

2. **Topic 14: Basic Algorithms**

   To ensure the application remains responsive during market hours, I prioritize algorithms with **logarithmic O(log n)** or **constant O(1)** time complexity for critical paths. For example, retrieving the current price of 'AAPL' would use a Hash Map lookup O(1) rather than iterating through a list of all 500 S&P companies O(n)

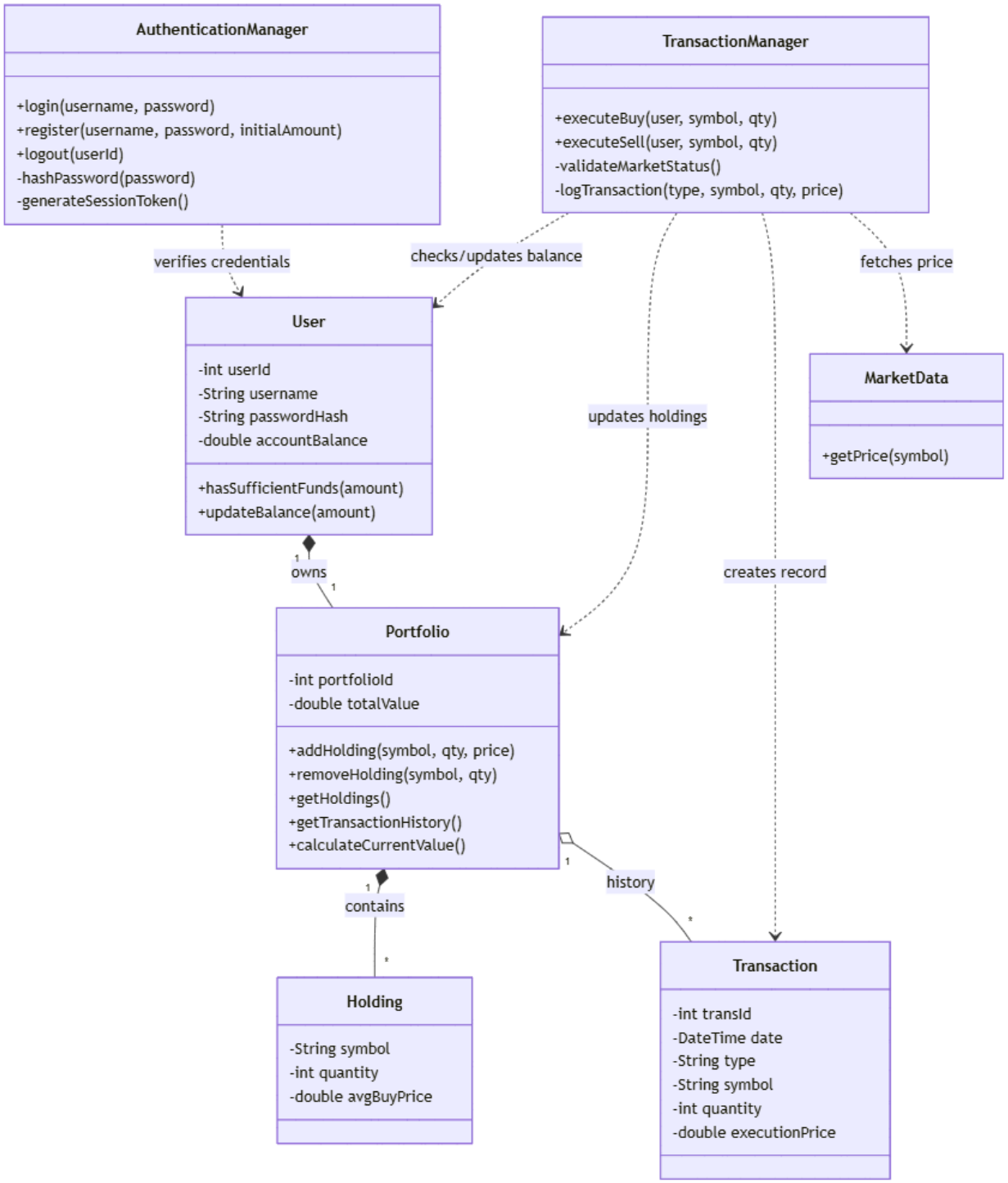
   The Portfolio Dashboard would use a comparison-based sort to organize holdings. When the user selects 'Sort by P/L', the system would apply a **Quicksort** algorithm (O(n log n) average case).

3. **Topic 20: Databases – Design and Query (The "System" Choice)**

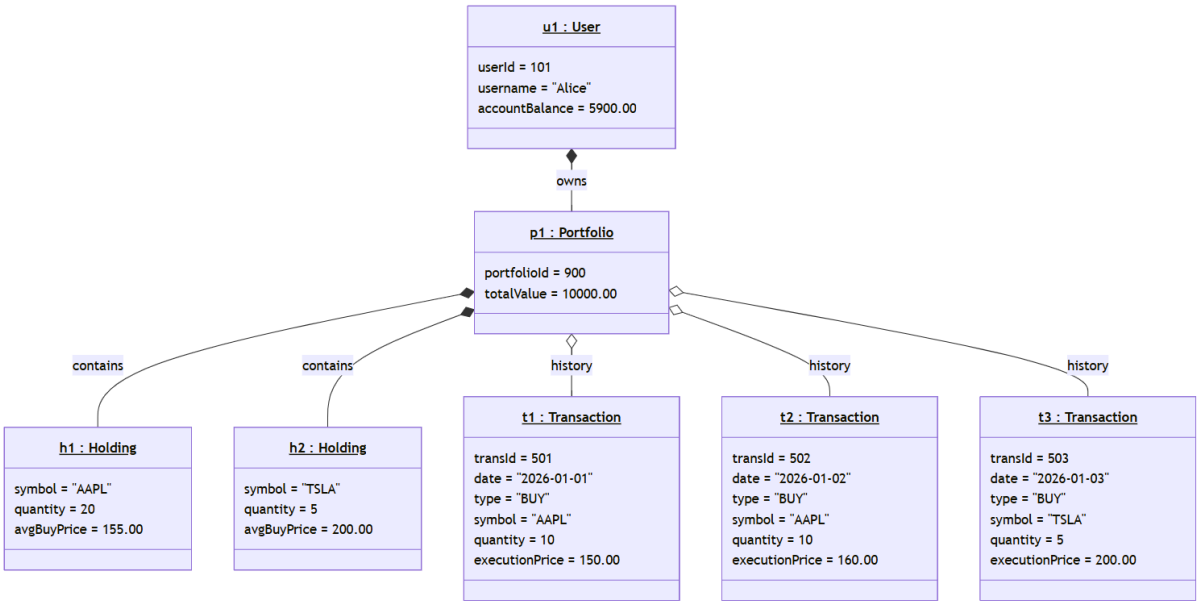
   Database optimization techniques are needed to ensure that as the number of users and historical trade records grows, the system remains responsive, allowing users to load complex portfolio charts instantly without lag. Furthermore, concurrency control mechanisms would guarantee data integrity; this ensures that even if thousands of users attempt to buy or sell stocks at the exact same moment, the database processes each transaction accurately and sequentially, preventing critical errors like negative balances or lost funds.

Diagrams

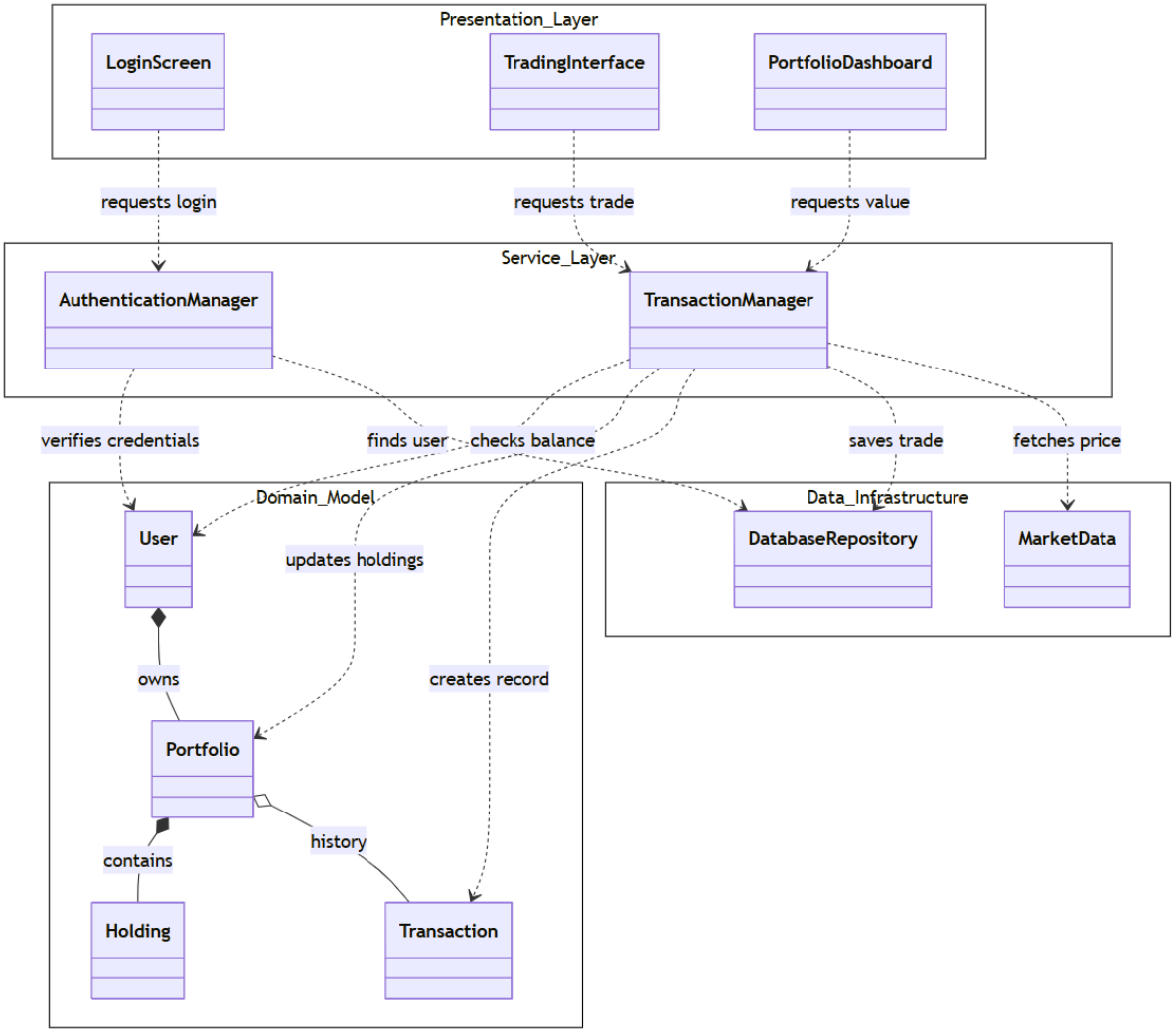
Class diagram



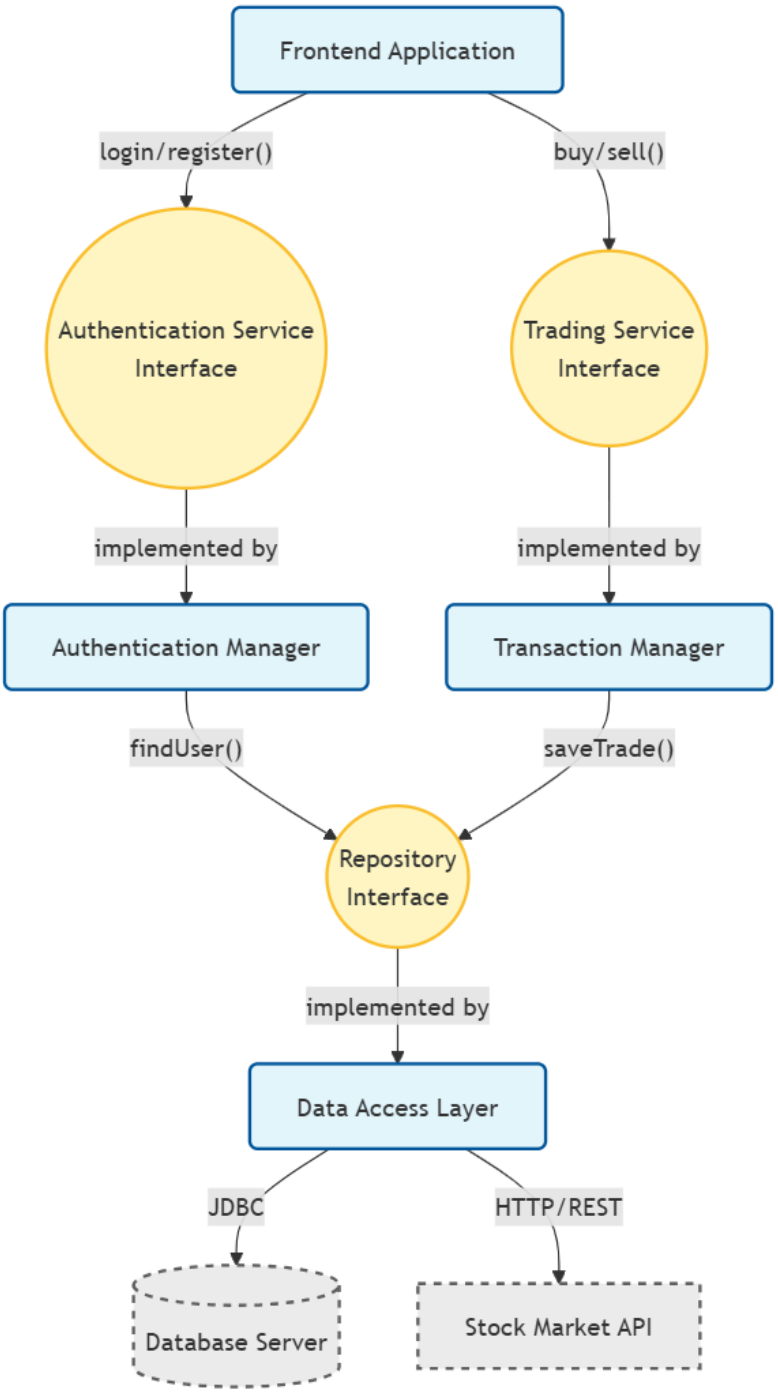
Object diagram



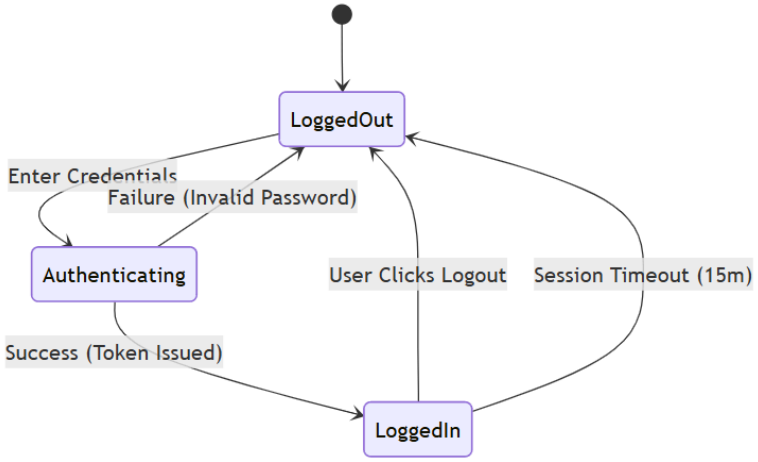
Package diagram



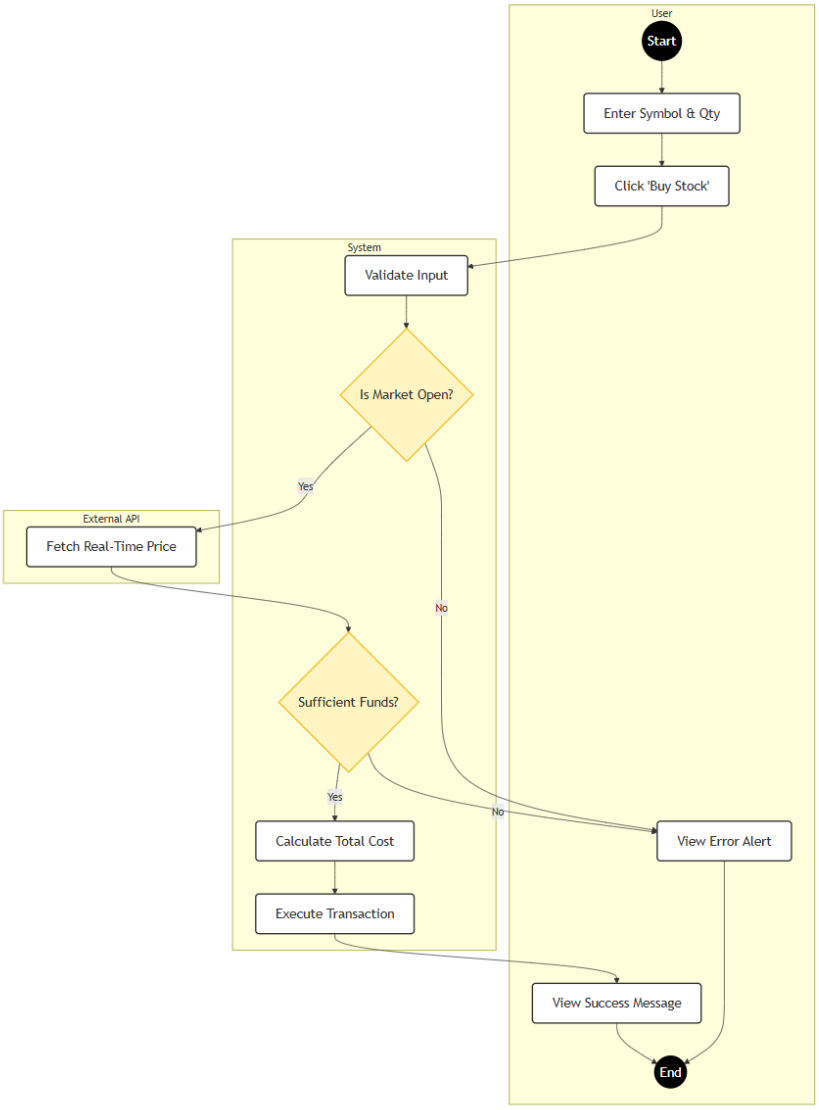
Component diagram



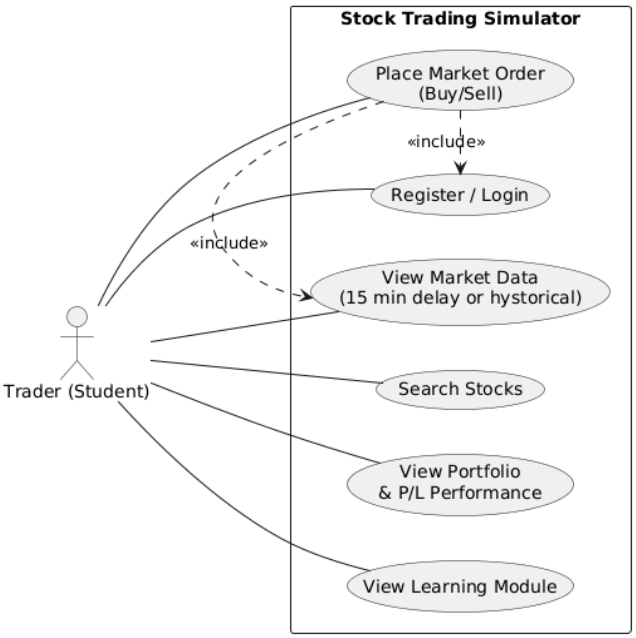
Application states diagram



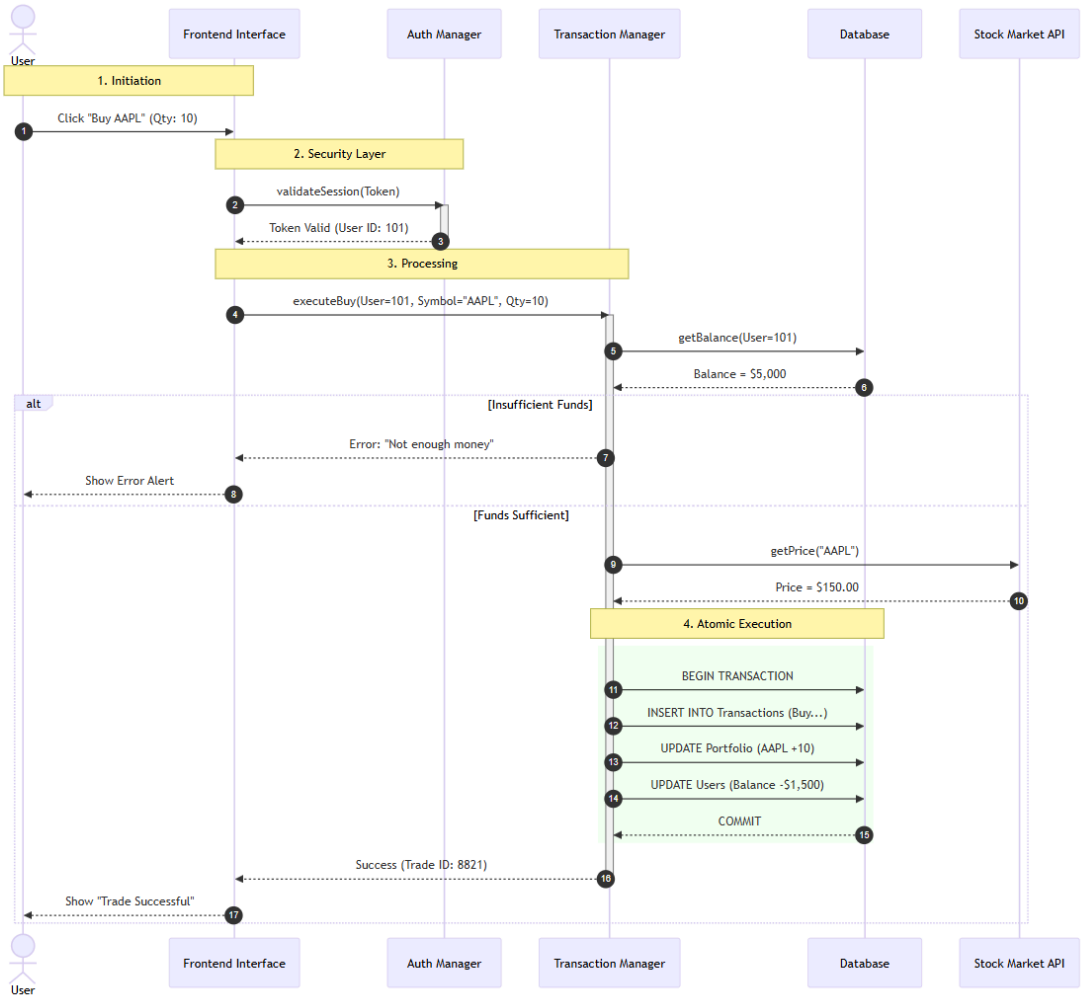
Buying Stocks states diagram



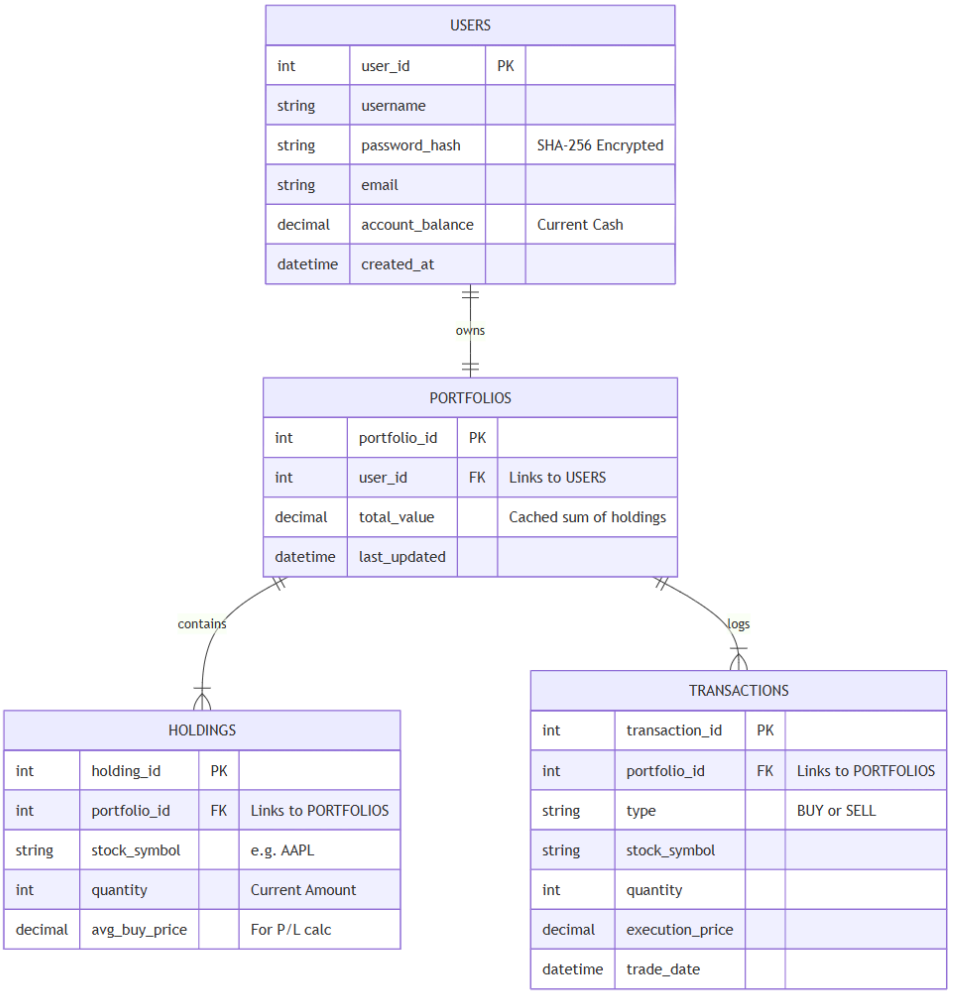
Use case diagram



Sequence diagram



ERD diagram


USERS
int user_id PK
string username
string password_hash SHA-256 Encrypted
string email
decimal account_balance Current Cash
datetime created_at
owns
PORTFOLIOS
int portfolio_id PK
int user_id FK Links to USERS
decimal total_value Cached sum of holdings
datetime last_updated
contains
logs
HOLDINGS
int holding_id PK
int portfolio_id FK Links to PORTFOLIOS
string stock_symbol e.g. AAPL
int quantity Current Amount
decimal avg_buy_price For P/L calc
TRANSACTIONS
int transaction_id PK
int portfolio_id FK Links to PORTFOLIOS
string type BUY or SELL
string stock_symbol
int quantity
decimal execution_price
datetime trade_date